

Attack–Fault Confounding at the Electronic Trading Gateway: Fragility under State Feedback, the Cost of Ambiguity over the Observation Horizon, and a Retrospective Study of Summary Matching

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Abstract

Trading-gateway operators must act on suspicious order flow without knowing whether a compromise or a software fault caused it. In a linear-Gaussian model of gateway-observable order flow, the causes are indistinguishable if and only if their projected injections coincide, and the ambiguity has a strictly positive cost; both facts specialise classical results. Once the incident feeds into the dynamics, confounding survives only if the injection difference is Kalman-unobservable: for almost every dynamics matrix, though not necessarily for structured market dynamics, only identical state injections, or identically distributed random ones, confound. For a pair confounded only at the output, the closed-form cost of ambiguity stays maximal at every horizon under output-level injection but falls below its maximum within at most n post-onset observations under state-level injection when (A, C) is observable. In four ABIDES designs in which a search tunes the fault toward a fixed compromise on one summary statistic, without a match to a pre-specified tolerance, no summary-matched regime reached chance on retrospective complete-episode features, also after multiplicity adjustment, in one design only narrowly. Each such cell, read as a classifier two-sample test, rejects equal observation laws. With a fault loss modelled from measured submitted volume, a classifier-based policy shows no supported deterioration against the best constant policy at this sample size under either

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fault-cost convention; against the best single-statistic threshold rule, 1 cell is a supported deterioration under the rebaselined convention, inconclusive after adjustment. Costs are illustrative reconstructions.

Keywords: cyber-physical security, electronic trading, attack–fault identifiability, cost of ambiguity, decision under uncertainty, incident response, agent-based market simulation

1. Introduction

A trading participant can generate harmful or suspicious orders because of a compromised credential, a software defect, its own order-generation policy, ordinary market conditions, or a combination of these. An operator — the incident-response controller of a broker or trading firm — must often act before the cause is known: throttle, cancel orders, revoke access, or isolate a process. Each action has a cost when applied to a case that did not need it, and each carries risk when withheld from a case that did. This paper answers two questions that come before when to act (timing is out of scope; Section 3.5).

First: **can the causes be told apart** from the evidence an operator is ordinarily permitted to observe (order acknowledgments, market data, session and process telemetry)? Second, if not: **how costly is that ambiguity**, measured against the best action an operator could choose without the observation? A threshold-based kill switch answers the second question without the first: it trips on an order rate, a cancel ratio or a loss limit and acts, with no attempt to establish cause.

This paper answers the first question with a formal identifiability result (Sections 3–4), scoped to a linear-Gaussian order-flow model, as is the closest structural analog, the literature distinguishing cyberattacks from physical faults in cyber-physical systems. The cost of ambiguity is defined with the loss table (Definition D7, Section 3.4); Sections 5–6 check the theory and estimate that cost in a linear pilot. Section 7 studies retrospectively, in the ABIDES agent-based market simulator, what matching two incident mechanisms on a chosen summary statistic does to classification and to decision cost (Section 7.1).

The security message is this. Exact attack/fault confounding requires the projected injections to coincide (Theorem 1). With few observed channels this is easy to meet under output-level injection: with a free fault sever-

ity it needs only $Cb_K \in \text{span}(Cb_F)$ (Corollary 1), which holds for every pair with $Cb_F \neq 0$ when one channel is observed. Once the incident’s actions feed back into the book, it survives only for injection differences in the unobservable subspace, and for almost every dynamics matrix only as the trivial $\theta_K b_K = \theta_F b_F$ (Theorem 2, Corollary 3); and, in every tested design, a calibration search that tuned the fault toward the compromise on a chosen summary statistic, without reaching a pre-specified match, did not bring them to chance.

Contributions.

1. The cost of ambiguity over the first H post-onset observations, in closed form (Proposition 4): for a pair Theorem 1 confounds whose state injections differ, it stays at its maximum at every horizon under output-level injection, and under state-level injection with observable dynamics it falls below it from an exact horizon $h^* \leq n$. It rests on classical observability applied to the compromise/fault pair: output-level confounding survives state-level injection only if the injection difference lies in the Kalman unobservable subspace [1, 2], so for almost every dynamics matrix only $\theta_K b_K = \theta_F b_F$ survives (Theorems 2–3 and Corollary 3, which use the attack-free special case of the object of [3]); Theorem 4 extends this to random injections of arbitrary law. In a simulation, an exactly confounded pair becomes classifiable under one stable, coupled, observable A while it stays at chance under output-level injection.
2. An ABIDES study of summary matching (Section 7). In the four tested injection designs, a calibration search that tunes the fault toward a fixed compromise, without matching it to a pre-specified tolerance, did not bring the summary-matched regime to chance, marginally so in D2, and in some designs the summary-matched regime was more separable than the contrast regime.

Section 7.5 states, as an outlook, the question these cells leave open.

2. Related work

Decision-theoretic cyber response. POMDP formulations of cyber defense choose actions under noisy alerts as an attacker progresses [4]; related work uses Stackelberg games over attack-response trees [5], balances service continuity against intrusion risk [6] (with learning-based extensions [7, 8]),

or supports decisions with models [9]. None of it prices market impact; our loss table uses a market-impact proxy, not a model of liquidity replacement.

Cyber-physical fault-versus-attack differentiation. The closest analog separates physical faults from cyberattacks; its classical core is fault diagnosis: two failure modes are separable when their output signatures differ [10], and in stochastic linear models fault-mode distinguishability is a divergence between output laws, zero exactly when they coincide [11]; Theorem 1 is this criterion for one compromise and one fault instance. In secure control, a divergence-based stealth bound limits every detector [12]. Pasqualetti, Dörfler and Bullo [3] characterize which attacks are undetectable or unidentifiable by an output-based monitor (those that excite only the zero dynamics); Theorem 2 uses the attack-free special case of this object, the Kalman unobservable subspace. Roy and Dey [13] derive, with an unknown-input observer, a closed-form fault/attack distinguishability criterion whose fault-mimicking condition is the analog of our Corollary 1. Observer-based fault and attack isolation schemes [14, 15] address the design side, and two recent designs use the closed loop itself: Zhou et al. [16] pair a plant-side fault detector with an observer-based attack detector, because multiplicative faults and additive attacks act differently on the closed-loop dynamics, and Xue et al. [17] add a plant-side check of the controller’s input–output consistency, which kernel (zero-dynamics, covert, replay) attacks excite and faults do not. These synthesise detectors for given fault and attack classes, and none prices the response or treats trading. Theorem 2 and Corollary 3 synthesise nothing: they fix one compromise/fault pair, take the dynamics matrix as generic, and show that, unless the two state injections are identical, confounding at the output survives state-level injection only on a Lebesgue-null set of dynamics matrices. In active fault diagnosis, inputs are designed over a finite horizon to minimise an upper bound on the probability of choosing the wrong linear model [18, 19], and Bayesian diagnosability is a windowed separation of predicted outputs whose growth drives identification [20]. Proposition 4 designs no input and uses no bound: for two Gaussian hypotheses on a passive window it gives the exact cost of ambiguity at every horizon in closed form, and the exact horizon $h^* \leq n$ from which, under state-level injection with (A, C) observable, it falls below its maximum; under output-level injection it never does. Data-driven work separates simulated faults from attacks with a classifier [21] or by fusing network and process data [22]. Closest to our framing, the triage framework of Ma et al. for energy systems attributes anomalies to physical faults or to measurement-integrity loss, prices misattribution

against blanket responses, and observes that exactly matched fault/attack pairs are discriminated at chance [23]. It derives no necessary-and-sufficient indistinguishability condition, cost floor or state-feedback fragility, and does not address trading. Its matched pairs are the empirical counterpart of our Theorem 1 and Proposition 1; Section 7 matches only summary statistics.

Linear stealth need not survive nonlinearity: stealthy injections must be redesigned for nonlinear grid models [24], and on Lie groups an attack keeps impact and stealth across operating conditions only if it commutes with the flow [25]. No analogous condition is known for our order-flow model; we claim neither that it holds nor that it fails.

Market manipulation detection and trading kill switches. A large literature detects manipulative patterns (spoofing, wash trading) from trade and order-book data [26]; it neither separates a compromised gateway from a faulty one nor prices the response. Compromised brokerage accounts can automate manipulative order generation [27], which supports modelling a compromise as a structured perturbation of order flow. A threshold trigger on an order-flow statistic is an observation-dependent policy, not one of this paper’s constant reference policies (Section 9).

Costly information acquisition. Work on asset markets [28] concerns traders’ incentives to learn about asset value. Closer in method, optimal stopping with costly sequential information studies when to stop paying for signals before a reversible or irreversible decision [29]. This paper buys no information: it bounds what passive observation of order flow can achieve, and leaves an action on a second channel to separate work (Section 7.5).

Scope of the contribution. Theorem 1 specialises classical results on the distinguishability of fault modes [10, 11], and Propositions 1–3 the Bayes risk of a two-point test [30, 31], to the gateway’s loss table; they fix notation and the cost floor, and are not the contribution. What we add is (1) the cost of ambiguity over the observation horizon in closed form, with the exact horizon $h^* \leq n$ from which state-level injection lowers it (Proposition 4), resting on a comparison of output-level and state-level injection of the same pair (Theorem 2, Corollary 3) that applies classical observability and extends to random injections (Theorem 4); (2) the compromise-versus-fault question at a trading gateway, with a market-impact loss table rather than a detection-accuracy metric; and (3) a retrospective simulator study of whether tuning the fault to match the compromise on a chosen summary statistic brings them to chance. Methodologically, the classification test is a classifier two-sample test [32], and the cost comparison is an expected-cost comparison

with constant policies over a prior grid.

3. Model

Throughout, K (“compromise”) and F (“fault”) name fully specified perturbations of the order-flow process — a directional bias, an elevated order rate or an altered cancellation pattern, with a chosen severity and onset — not claims about real attackers or real defects. The results concern *behavioural-mechanism identification*, whether two specified generative processes can be told apart from gateway-observable order flow, not inference about intent or authorization, which would need other evidence (credential logs, code-change records).

3.1. Order-flow dynamics

Let $x_t \in \mathbb{R}^n$ be a latent representation of normal order-flow state (for example, components loading onto price pressure, volume intensity and cancellation propensity), evolving under known linear-Gaussian dynamics with transition matrix A and process-noise covariance Q . The complete model is Definition D2 below.

3.2. Incident injection

At an onset time τ , possibly unknown to the operator, an incident of mechanism $M \in \{K, F\}$ injects an additive perturbation into the pre-observation order-flow signal, with direction b_M and severity θ_M .

Definition D1 (Mechanism instance). A mechanism instance is a triple (M, θ_M, b_M) with cause $M \in \{K, F\}$ (K : compromise, F : fault), a deterministic severity $\theta_M \in \mathbb{R}$ and a deterministic injection direction $b_M \in \mathbb{R}^n$. Statements compare one fixed instance of K with one fixed instance of F (pairwise-instance reading); no distribution over severities, directions or onsets within a cause is modelled.

3.3. Gateway-observable output

The operator observes a noisy projection of order flow, typically lower-dimensional ($m < n$), through a fixed, known matrix $C \in \mathbb{R}^{m \times n}$, representing what a gateway or market-data feed surfaces (spread, depth, cancellation rate). In this model the incident shifts the signal at the point of observation only; Section 4.1 lets it propagate through the dynamics.

Definition D2 (Output-level injection model). Fix n, m , a matrix $A \in \mathbb{R}^{n \times n}$, an observation matrix $C \in \mathbb{R}^{m \times n}$, covariance matrices $Q \in \mathbb{R}^{n \times n}$ and $\Sigma \in \mathbb{R}^{m \times m}$, and an onset time τ . Under a mechanism instance (M, θ_M, b_M) (Definition D1), the latent state, the perturbed state and the observation evolve for $t \geq 0$ as

$$\begin{aligned} x_{t+1} &= Ax_t + w_t, & \tilde{x}_t &= x_t + \mathbb{I}[t \geq \tau] \theta_M b_M, \\ y_t &= C\tilde{x}_t + \varepsilon_t, \end{aligned}$$

with $w_t \sim \mathcal{N}(0, Q)$ and $\varepsilon_t \sim \mathcal{N}(0, \Sigma)$. The injection enters the observation only; the state recursion does not see it.

Everything except the injection term is common to the two causes. Section 4 rests on the following standing assumptions, which the implementation satisfies; the finite-horizon experiments of Sections 5–6 always include time steps after the onset, so (A5) holds.

Standing assumptions.

(A1) Gaussian, independent noise. $w_t \sim \mathcal{N}(0, Q)$ and $\varepsilon_t \sim \mathcal{N}(0, \Sigma)$, with Q and Σ symmetric positive semidefinite; the family $\{w_t\}_{t \geq 0} \cup \{\varepsilon_t\}_{t \geq 0}$ is mutually independent (across time and across the two sequences), and its law does not depend on the instance.

(A2) Deterministic initial state. $x_0 = 0$. (The proofs use only that x_0 is Gaussian, possibly degenerate, independent of the noise, with a law that does not depend on the instance; $x_0 = 0$ is the implemented special case.)

(A3) Known, common model. A, Q, C, Σ are known and identical under both causes; only the injection term $\theta_M b_M$ depends on the instance.

(A4) Common onset. The onset τ is deterministic, common to both causes, not necessarily known to the operator. (The proofs use only that τ is deterministic and common; no statement requires a policy to know τ .)

(A5) Window reaches the onset. The observation window W contains at least one $t \geq \tau$. Theorem 2 uses the stronger form $W \supseteq \{t : t \geq \tau\}$ and Theorem 3 the stronger form $W \supseteq \{\tau, \dots, \tau + H - 1\}$.

(A6) Positive-definite observation noise. $\Sigma \succ 0$. Used only to apply Proposition 3 to the model on a finite window: it makes the covariance of the stacked observations positive definite ($V \succ 0$). No

other statement uses a positivity condition on Σ or Q ; Q is only required to be positive semidefinite.

(A7) Ordered loss table. $H_K > C_K$ and $C_F > H_F$ (Definition D6): leaving a compromise alone costs strictly more than containing it, and containing a fault costs strictly more than leaving it alone. Propositions 1 and 2 carry it as their hypotheses $\text{HARM} > C_{OP}$ and $\text{DISRUPT} > 0$; Propositions 3 and 4 as $a, b > 0$. No identifiability statement (Theorems 1–4, Corollaries 1–3) uses it. Every use of the threshold $q^* = (C_F - H_F)/[(H_K - C_K) + (C_F - H_F)]$, of $R_{\text{oracle}} = p C_K + (1 - p) H_F$ and of the best constant policy’s excess $\min\{p(H_K - C_K), (1 - p)(C_F - H_F)\}$ over the oracle requires it, so an empirical loss table to which these are applied must satisfy it in every cell.

Definition D3 (Observation window; observational indistinguishability). An observation window is a set $W \subseteq \{0, 1, 2, \dots\}$ of time indices, finite or not; the observation trajectory on W is $Y_W = (y_t)_{t \in W}$. Write P_M^W for the law of Y_W under instance (M, θ_M, b_M) (for infinite W , the law on the product σ -algebra, determined by its finite-dimensional marginals). Instances (K, θ_K, b_K) and (F, θ_F, b_F) are observationally indistinguishable on W if $P_K^W = P_F^W$. Theorem 1 uses $W = \{t : t \geq 0\}$; “indistinguishable at every $t \geq \tau$ ” (Theorem 2) means $P_K^W = P_F^W$ for $W = \{t : t \geq \tau\}$; Theorem 3 (the finite-horizon version of Theorem 2) uses $W = \{\tau, \dots, \tau + H - 1\}$.

Definition D4 (Distinguishable in the population limit). Two instances are distinguishable in the population limit on W if they are not observationally indistinguishable on W , i.e. $P_K^W \neq P_F^W$ (equivalently $\text{TV}(P_K^W, P_F^W) > 0$, Definition D8). This is a statement about the two laws only; it asserts nothing about the accuracy attainable from finitely many observations.

Definition D5 (Gaussian discriminability). If $Y \mid M \sim \mathcal{N}(\mu_M, V)$, $M \in \{K, F\}$, with a common covariance $V \succ 0$, let $\delta = \mu_K - \mu_F$ and $d = \sqrt{\delta^\top V^{-1} \delta} \geq 0$ (the whitened mean separation).

3.4. The loss table, and what it assumes

The cost analysis uses the two-action, two-cause loss table

Action	Compromise K	Fault F
Contain	C_{OP}	DISRUPT
Leave alone	HARM	0

a special case of the general table below; Proposition 1 keeps these symbols.

Definition D6 (Loss table). The general two-action, two-cause loss table $L(a, M)$, $a \in \{\text{contain, leave alone}\}$, $M \in \{K, F\}$, is

$$\begin{aligned} L(\text{contain}, K) &= C_K, & L(\text{contain}, F) &= C_F, \\ L(\text{leave alone}, K) &= H_K, & L(\text{leave alone}, F) &= H_F, \end{aligned}$$

with $C_K, C_F, H_K, H_F \in \mathbb{R}$. This paper's table (the two-action table printed above this definition) is the special case

$$C_{OP} = C_K, \quad \text{DISRUPT} = C_F, \quad \text{HARM} = H_K, \quad H_F = 0.$$

Propositions 1 and 2 are stated in this paper's symbols C_{OP} , DISRUPT, HARM.

Definition D7 (Prior, policies, Bayes risk, oracle risk, cost of ambiguity). The cause is drawn as $M = K$ with prior probability $p = P(K)$ and $M = F$ with probability $1 - p$; given M , the observation trajectory Y_W has law P_M^W . A policy is a measurable map π from observation trajectories on W to $[0, 1]$, read as the probability of the action *contain* (randomized policies allowed; a deterministic policy takes values in $\{0, 1\}$). Its expected cost is

$$\begin{aligned} R(\pi) &= p \mathbb{E}_K [\pi(Y_W) C_K + (1 - \pi(Y_W)) H_K] \\ &\quad + (1 - p) \mathbb{E}_F [\pi(Y_W) C_F + (1 - \pi(Y_W)) H_F]. \end{aligned}$$

The best achievable expected cost is $R^* = \inf_{\pi} R(\pi)$; the oracle, which observes M , has expected cost $R_{\text{oracle}} = p \min\{C_K, H_K\} + (1 - p) \min\{C_F, H_F\}$; the cost of ambiguity is $\text{CoA}^* = R^* - R_{\text{oracle}}$. Where the observation is the raw trajectory Y the subscript Y is written (R_Y^* , CoA_Y^*).

Definition D8 (Observation densities; total variation). Let μ be a σ -finite measure dominating P_K and P_F (e.g. the finite measure $P_K + P_F$), and f_K, f_F their densities with respect to μ ; $\int \cdot dy$ denotes integration against μ . The total variation distance is

$$\text{TV}(P, P') = \sup_B |P(B) - P'(B)| = \frac{1}{2} \int |f - f'| dy,$$

so that $1 - \text{TV}(P_K, P_F) = \int \min\{f_K, f_F\} dy$.

The table encodes a scenario in which leaving a fault alone is free, for example a logging error that inflates apparent order counts. The strict value of distinguishing the two causes is zero whenever one action is optimal for

both; Proposition 1’s cost floor is a property of loss structures in which the causes call for different actions. The simulated fault mechanisms of Section 7 submit real orders, so Section 6.2 replaces the zero entry by a positive fault loss H_F , modelled from the submitted volume measured on the simulated episodes. The per-design audit of the fault mechanisms is in Supplement S4.

3.5. Observation timing and scope

Theorem 1 and Proposition 1 concern the whole observation trajectory, and both the linear pilot and the ABIDES study use whole-episode summary features (mean, standard deviation and net drift of each observed channel), not a prefix available at a pre-committed decision deadline. The empirical study is therefore retrospective diagnosis: a real-time claim would need an explicit observation cutoff and prefix features, and we make none. The identifiability results, which concern the laws of whole trajectories, are unaffected, but the costs of Sections 6–7 do not measure the value of information at an operational decision point.

4. Theoretical results

Theorem 1 and Propositions 1–3 specialise classical results to the gateway (Section 2); we state them because the rest of the paper uses their exact form. Proofs of Theorem 2, Corollary 3 and Proposition 4 are in Appendix A; Supplement S1, submitted with the manuscript, proves every numbered statement.

Theorem 1 (Indistinguishability criterion). Under the model of Section 3, with common onset time τ , mechanism instances (K, θ_K, b_K) and (F, θ_F, b_F) induce identical distributions over the observation trajectory $\{y_t\}_{t \geq 0}$ if and only if

$$\theta_K C b_K = \theta_F C b_F. \quad (*)$$

Theorem 1 uses Definition D2, Definition D1, Definition D3 (window $W = \{t : t \geq 0\}$) and assumptions (A1)–(A5).

No rank, observability or definiteness condition on A , C , Q or Σ is needed; the onset τ may be any deterministic time.

Corollary 1 (Fault-mimicking compromise). Fix $u = C b_K$, $v = C b_F$, and a nonzero compromise severity $\theta_K \neq 0$. If the fault severity θ_F may be any real number, then

$$\exists \theta_F \in \mathbb{R} : \theta_K u = \theta_F v \iff u \in \text{span}(v).$$

Corollary 1 uses Definition D1 and condition (*) of Theorem 1; the equivalence itself is linear algebra and uses no probabilistic assumption.

Two precisions. Linear dependence of u and v means rank at most one, and its subcases differ: if $v = 0$ but $u \neq 0$ no θ_F solves the equation, and if both vanish every θ_F does; with positive severities, nonzero u, v must lie on the same ray. Theorems 1–3 and Propositions 1–3 compare one fixed instance of each mechanism (Definition D1); Theorem 4 (Section 4.2) replaces instances by random injections drawn once per episode, and laws averaged over onsets would need a model this paper does not specify.

Corollary 2 (Separation implies distinguishability, not high accuracy). If $\text{rank}([Cb_K \ Cb_F]) = 2$, no pair (θ_K, θ_F) other than $(0, 0)$ satisfies (*); for any instance pair with $(\theta_K, \theta_F) \neq (0, 0)$, K and F are then distinguishable in the population limit.

Corollary 2 uses Definition D1, Definition D3, Definition D4 and, through Theorem 1, Definition D2 and assumptions (A1)–(A5).

Separation does not imply good finite-sample classification: the projected directions can be small relative to the noise. The benchmark is the whitened separation d of Definition D5; under equal priors the Bayes-optimal accuracy and area under the ROC curve (AUC) are the standard identities [33, 34]

$$\text{Acc}^* = \Phi(d/2), \quad \text{AUC}^* = \Phi(d/\sqrt{2}).$$

Proposition 1 (Cost of ambiguity is bounded below under confounding). Suppose (*) holds (the confounded case), $0 < p < 1$, DISRUPT > 0 , and HARM $> C_{OP}$ (harm strictly worse than routine containment). Then the best achievable expected cost over the policies of Definition D7 (randomized policies allowed) is

$$R^* = \min\{p C_{OP} + (1 - p) \text{DISRUPT}, \ p \text{HARM}\},$$

the oracle’s expected cost is $R_{\text{oracle}} = p C_{OP}$, and

$$\text{CoA}^* = R^* - R_{\text{oracle}} = \min\{(1 - p) \text{DISRUPT}, \ p(\text{HARM} - C_{OP})\} > 0.$$

Proposition 1 uses Definition D6 (this paper’s symbols), Definition D7 and, through Theorem 1 (to pass from (*) to $P_K^W = P_F^W$), Definition D2, Definition D1, Definition D3 and assumptions (A1)–(A5).

Exact equality is a brittle operational target, so we extend the bound to approximate confounding.

Proposition 2 (Total-variation extension of Proposition 1). Let the loss table and prior be as in Proposition 1, with $0 < p < 1$, $\text{DISRUPT} > 0$ and $\text{HARM} > C_{OP}$, but do not assume (*). Let f_K, f_F be the observation densities (Definition D8) and set $a = p(\text{HARM} - C_{OP})$, $b = (1 - p) \text{DISRUPT}$. Then

$$\text{CoA}^* = \int \min\{af_K(y), bf_F(y)\} dy = \frac{a + b - \int |af_K - bf_F| dy}{2},$$

$$\min(a, b) (1 - \text{TV}(P_K, P_F)) \leq \text{CoA}^* \leq \min(a, b),$$

and, if a common surrogate law $\bar{P}$ satisfies $\text{TV}(P_K, \bar{P}) \leq \epsilon_K$ and $\text{TV}(P_F, \bar{P}) \leq \epsilon_F$, then

$$\text{CoA}^* \geq \min(a, b) \max\{0, 1 - \epsilon_K - \epsilon_F\}.$$

Proposition 2 uses Definition D6 (this paper’s symbols), Definition D7 and Definition D8; it uses none of (A1)–(A6): it holds for any two observation laws.

The surrogate-law bound states the premise any “approximate confounding transfers approximately” argument needs: a bound on distributional approximation error, not a match on one statistic. A classifier performing near chance cannot certify a small total variation distance. The surrogate-law bound is the cost counterpart of ϵ -stealth in stochastic control [12]: an attacker whose observation law stays within a small distance of a law a fault can also produce keeps the cost of ambiguity near its maximum.

For the equal-covariance Gaussian setting, likelihood-ratio integration [31] gives the population-optimal ambiguity cost as a function of d , the unequal-prior cost analog of Corollary 2; a summary-based classifier can only do worse.

Proposition 3 (Gaussian unequal-prior ambiguity cost). Let $Y | M \sim \mathcal{N}(\mu_M, V)$, $M \in \{K, F\}$, with common covariance $V \succ 0$ and discriminability $d > 0$ (Definition D5), prior $p = P(K)$, and the general loss table with $a = p(H_K - C_K)$, $b = (1 - p)(C_F - H_F)$, $a, b > 0$. Then the population-optimal ambiguity cost is

$$\text{CoA}_Y^*(d) = a \Phi\left(\frac{\log(b/a)}{d} - \frac{d}{2}\right) + b \Phi\left(\frac{\log(a/b)}{d} - \frac{d}{2}\right),$$

which tends to $\min(a, b)$ as $d \rightarrow 0^+$ (the value at $d = 0$, exact confounding, recovering Proposition 1, is obtained as this limit) and decreases toward 0 as $d \rightarrow \infty$.

Proposition 3 uses Definition D6 (general symbols), Definition D7, Definition D8 and Definition D5; applied to the model on a finite window it needs (A1)–(A5) (common covariance) and (A6) ($V \succ 0$).

4.1. Confounding is fragile within the linear model: state-level injection

Theorem 1 models the incident as an output-level disturbance: it shifts x_t only at the point of observation, while the recursion $x_{t+1} = Ax_t + w_t$ evolves as if nothing happened. A compromise or a persistent fault behaves more like a state-level disturbance: behaviour at t feeds into the state at $t + 1$, as an incident agent’s orders feed into the book that other agents react to. This changes the confounding condition. The objects involved, the unobservable subspace and the signatures a monitor can separate, are those of geometric failure detection and of attack-space analysis in secure control [10, 35]; what we add is the comparison of the same attack/fault pair under the two injection levels.

Definition D9 (State-level injection model). As Definition D2, except that the perturbed state drives the recursion from the onset on:

$$\begin{aligned} x_{t+1} &= Ax_t + w_t \quad (t < \tau), & x_{t+1} &= A\tilde{x}_t + w_t \quad (t \geq \tau), \\ \tilde{x}_t &= x_t + \mathbb{I}[t \geq \tau] \theta_M b_M, & y_t &= C\tilde{x}_t + \varepsilon_t. \end{aligned}$$

The injection is persistent (added at every $t \geq \tau$) and propagates through A .

Definition D10 (Unobservable subspace; observability). The (Kalman) unobservable subspace of (A, C) is $\mathcal{N}(A, C) = \bigcap_{k=0}^{n-1} \ker(CA^k)$. The pair (A, C) is observable if $\mathcal{N}(A, C) = \{0\}$, equivalently if the stacked matrix

$$\begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix}$$

has rank n .

Theorem 2 (Confounding under state-level injection requires unobservability). Under the state-level injection model, let $\Delta_0 = \theta_K b_K - \theta_F b_F$. Mechanism instances (K, θ_K, b_K) and (F, θ_F, b_F) are observationally

indistinguishable at every $t \geq \tau$ if and only if Δ_0 lies in the unobservable subspace of the pair (A, C) , i.e.

$$\Delta_0 \in \bigcap_{k=0}^{n-1} \ker(CA^k),$$

the classical (Kalman) unobservable subspace of (A, C) , where n is the state dimension.

Theorem 2 uses Definition D9, Definition D1, Definition D3 (window $W = \{t : t \geq \tau\}$), Definition D10 and assumptions (A1)–(A4) with (A5) in its all-time form.

Theorem 3 (Finite-horizon version of Theorem 2). Under the state-level injection model, let $\Delta_0 = \theta_K b_K - \theta_F b_F$, let $H \geq 1$ be the number of post-onset observations, and define

$$G_H = \begin{bmatrix} C \\ C(I + A) \\ \vdots \\ C \sum_{j=0}^{H-1} A^j \end{bmatrix}.$$

Mechanism instances (K, θ_K, b_K) and (F, θ_F, b_F) induce the same law of the observations over the window of H post-onset observations if and only if $G_H \Delta_0 = 0$, which holds if and only if $CA^j \Delta_0 = 0$ for $j = 0, \dots, H-1$. If $H \geq n$, this is equivalent to the all-time condition of Theorem 2, $\Delta_0 \in \bigcap_{k=0}^{n-1} \ker(CA^k)$.

Theorem 3 uses Definition D9, Definition D1, Definition D3 (window $W = \{\tau, \dots, \tau + H - 1\}$), Definition D10 and assumptions (A1)–(A4) with (A5) in its finite-horizon form.

The result is not inherently asymptotic: for $H \geq n$, exact equality over H post-onset observations already forces the all-time Kalman condition. What depends on horizon and noise is discriminability when the condition fails, which Theorem 2 does not bound; Proposition 4 (Section 4.2) gives its population cost.

Corollary 3 (Confounding under state-level injection with observable dynamics). Under the hypotheses of Theorem 2, if (A, C) is observable, then mechanism instances (K, θ_K, b_K) and (F, θ_F, b_F) are observationally indistinguishable at every $t \geq \tau$ if and only if $\theta_K b_K = \theta_F b_F$.

Corollary 3 uses everything Theorem 2 uses, and Definition D10; its accompanying remark also uses Definition D11.

Definition D11 (Generic). For fixed C , a property of A holds generically if it holds for Lebesgue-almost every $A \in \mathbb{R}^{n \times n}$, i.e. it fails only on a Lebesgue-null subset of $\mathbb{R}^{n \times n}$.

Remark (Theorem 1’s confounded pairs are generically fragile under state-level injection). Theorem 1 only requires $\Delta_0 \in \ker(C)$ — a single condition. Theorem 2 requires Δ_0 to survive n such conditions simultaneously ($\ker(C), \ker(CA), \dots, \ker(CA^{n-1})$). Whenever (A, C) is observable (for fixed $C \neq 0$, the generic case for a randomly chosen A), the unobservable subspace is $\{0\}$, so **the only confounding robust to state-level injection is the trivial one where $\theta_K b_K = \theta_F b_F$ exactly as vectors** — i.e., the two mechanisms must push the identical linear combination of latent state variables, not merely produce the identical *projection* under C .

The pilot’s construction (Section 5), which uses the null space of C to make $b_K \neq b_F$ with $Cb_K = Cb_F$, is exactly such a pair: $\Delta_0 \in \ker(C)$ but $\theta_K b_K \neq \theta_F b_F$. By Corollary 3 it cannot remain confounded under state-level injection whenever (A, C) is observable. The pilot’s own A , a scalar multiple of the identity, is the measure-zero exception, so we re-ran the pilot with a stable, coupled and observable A , 10 seeds per magnitude, each with 480 balanced test episodes (a constant prediction scores one half). Mean test accuracy over seeds under state-level injection rises from 0.628 at the smallest of 16 magnitudes (95% interval over seeds 0.611–0.645) to 1.000; under output-level injection with the same dynamics it stays between 0.497 and 0.503, and 0 of its intervals exclude one half (Supplement S2). With the pilot’s scalar A , 1 of the 16 state-level intervals excludes one half (smallest magnitude, 0.501–0.522), about as many as unadjusted intervals on shared seeds give by chance.

A qualification: “almost every” concerns a full-measure set of matrices; real, structured market dynamics can have unobservable components by construction, and we have not established that they behave like a generic matrix. Supplement S2 gives three further qualifications (finite-horizon discriminability; identical state effects with different meanings; persistence versus state-level injection).

Theorem 2’s mechanism is not isolated in the simulator (Section 7.4).

4.2. Random injections and the observation horizon

Theorems 1–3 compare one fixed instance of each cause, whereas an attacker chooses severity and direction from a class. Theorem 4 lets each cause

inject a random vector, drawn once per episode; the instances of Definition D1 are its point-mass case.

Theorem 4 (Class-level identifiability). Let each cause $M \in \{K, F\}$ inject a random vector $s_M \in \mathbb{R}^n$ in place of $\theta_M b_M$, drawn once per episode and persistent (the same draw acts at every $t \geq \tau$), with an arbitrary law, and independent of the pair $(x_0, (w_t, \varepsilon_t)_{t \geq 0})$ jointly (independence of x_0 and of the noise family separately is not enough) (a mechanism instance of Definition D1 is the point-mass case $s_M = \theta_M b_M$). Write $\mathcal{O}_H = [C^\top (CA)^\top \dots (CA^{H-1})^\top]^\top$.

- (i) Under the output-level injection model (Definition D2), on any window W that contains at least one $t \geq \tau$, K and F induce the same law of Y_W if and only if Cs_K and Cs_F have the same law on $\mathbb{R}^m$.
- (ii) Under the state-level injection model (Definition D9), on the window $W = \{\tau, \dots, \tau + H - 1\}$, they induce the same law of Y_W if and only if $\mathcal{O}_H s_K$ and $\mathcal{O}_H s_F$ have the same law; for $H \geq n$, and for $W = \{t : t \geq \tau\}$, if and only if $\mathcal{O}_n s_K$ and $\mathcal{O}_n s_F$ have the same law.
- (iii) If (A, C) is observable (Definition D10), state-level indistinguishability on $W = \{t : t \geq \tau\}$ holds if and only if s_K and s_F have the same law.

Theorem 4 uses Definition D2 and Definition D9 with s_M in place of $\theta_M b_M$, Definition D3, Definition D10 and assumptions (A1)–(A5) (the window forms stated in (i)–(iii)); it adds the three hypotheses on s_M stated in the theorem (one draw per episode, persistence, joint independence), and uses no positivity condition.

Part (iii) is Corollary 3 for the laws of attacker and fault injections rather than for instances: with (A, C) observable, only identically distributed injections confound under state-level injection. Theorem 3 says when exact confounding fails over H post-onset observations; Proposition 4 gives what the remaining ambiguity costs as a function of H .

Proposition 4 (Cost of ambiguity over the observation horizon).

Fix mechanism instances (K, θ_K, b_K) and (F, θ_F, b_F) , $\Delta_0 = \theta_K b_K - \theta_F b_F$, the general loss table with $a = p(H_K - C_K) > 0$ and $b = (1 - p)(C_F - H_F) > 0$, and for $H \geq 1$ the window $W_H = \{\tau, \dots, \tau + H - 1\}$ of the first H post-onset observations. Let V_H be the covariance of Y_{W_H} (it is the same under both causes and under both injection models), let G_H be as in Theorem 3, and set

$$d_H^{\text{st}} = \|V_H^{-1/2} G_H \Delta_0\| \quad (\text{state level, Definition D9}), \quad d_H^{\text{out}} = \|V_H^{-1/2} (\mathbf{1}_H \otimes C \Delta_0)\| \quad (\text{output level, Definition D2})$$

Under either model, with d_H the corresponding value, the cost of ambiguity

on W_H is $\text{CoA}_H^* = \text{CoA}_Y^*(d_H)$, the closed form of Proposition 3, if $d_H > 0$, and $\text{CoA}_H^* = \min(a, b)$ if $d_H = 0$. Moreover:

- (i) under either model, $H \mapsto \text{CoA}_H^*$ is non-increasing;
- (ii) under output-level injection, $\text{CoA}_H^* = \min(a, b)$ for every H if $C\Delta_0 = 0$ (condition $(*)$ of Theorem 1), and $\text{CoA}_H^* < \min(a, b)$ for every H otherwise;
- (iii) under state-level injection, $\text{CoA}_H^* < \min(a, b)$ if and only if $CA^j\Delta_0 \neq 0$ for some $j < H$; if (A, C) is observable and $\Delta_0 \neq 0$, this holds for every $H \geq h^*$, where $h^* = 1 + \min\{j : CA^j\Delta_0 \neq 0\} \leq n$.

In particular, for a pair that Theorem 1 confounds ($C\Delta_0 = 0$) with $\Delta_0 \neq 0$ and (A, C) observable, the cost of ambiguity stays at $\min(a, b)$ at every horizon under output-level injection, and falls strictly below it from horizon $h^* \in \{2, \dots, n\}$ on under state-level injection.

Proposition 4 uses Definition D2, Definition D9, Definition D1, Definition D3 (window W_H), Definition D6 (general symbols), Definition D7, Definition D5, Definition D10, Proposition 3, Theorem 3 and assumptions (A1)–(A4) and (A6) ((A5) holds by construction of W_H).

Proposition 4 turns the fragility of Section 4.1 into a cost-versus-horizon curve: under output-level injection a confounded pair with $\Delta_0 \neq 0$ costs $\min(a, b)$ however long the operator observes, while under state-level injection with (A, C) observable it falls strictly below $\min(a, b)$ from a horizon $h^* \leq n$ on. Figure 1 evaluates it for the pilot’s confounded pair with the coupled observable A of the Theorem 2 rerun: the output-level cost stays at $\min(a, b)$ and the state-level cost falls below it from $h^* = 2$ on, while with the pilot’s own scalar dynamics, the measure-zero exception above, both stay at $\min(a, b)$. It is a population result on a fixed window (Section 3.5).

5. Pilot-scale validation in the linear model

We instantiate the model at small scale (a few latent dimensions, fewer observed channels, AR(1) dynamics; parameters in Supplement S2) as a check that the theory is internally consistent and detectable in simulation, not as a market-realistic study. Two regimes are built by the choice of injection directions: *confounded*, with b_K, b_F chosen through the null space of C so that $Cb_K = Cb_F$ exactly, satisfying Theorem 1’s condition; and *separated*, with Cb_K, Cb_F linearly independent. A logistic-regression classifier is trained on

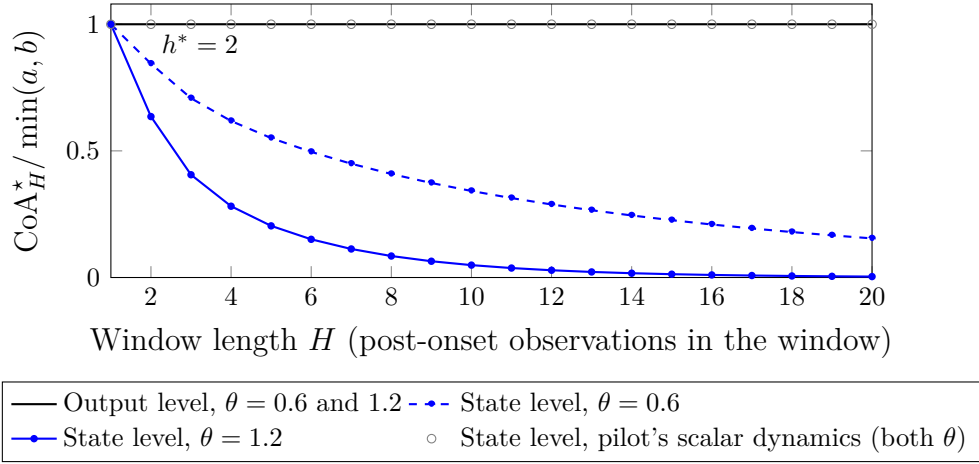


Figure 1: Population cost of ambiguity on the window of the first H post-onset observations (Proposition 4), as a fraction of $\min(a, b)$, for the pilot’s confounded pair ($Cb_K = Cb_F$) at prior $p = 0.5$. The pilot’s C , Q , Σ , τ and injection directions are used with the coupled observable A of the Theorem 2 rerun (Section 4.1), not with the pilot’s own dynamics. Under output-level injection the cost stays at $\min(a, b)$ at both magnitudes $\theta = 0.6$ and $\theta = 1.2$; under state-level injection it falls below $\min(a, b)$ from $h^* = 2 \leq n = 3$ on. With the pilot’s own scalar dynamics, which are not observable with the pilot’s C (observability rank $2 < n$), the state-level cost also stays at $\min(a, b)$ at every H (grey markers on the output-level line), the exception of Section 4.1. Every prior of the toy grid is in the results file.

whole-episode summary statistics of each observed channel (mean, standard deviation, net drift), in both regimes and at two injection magnitudes.

As Theorem 1 predicts, the confounded regime’s accuracy interval contains one half at both injection magnitudes: a more severe incident is not easier to classify when the two observation laws are identical. In the separated regime accuracy is high and improves with magnitude, as Corollary 2 predicts through the whitened separation. Three further classifier families (a random forest, Gaussian naive Bayes and a single-feature threshold rule) give the same pattern (Supplement S2), as Theorem 1 implies for any passive classifier.

6. Estimating the cost of ambiguity: threshold, loss entries and the linear pilot

Terminology. With the oracle and best observation-based risks R_{oracle} , R_Y^* of Definition D7, the best constant action $R_{\text{const}} = \min_a \mathbb{E}[L(a, M)]$ and

Table 1: Linear pilot: held-out accuracy and AUC by regime and injection magnitude (480 test episodes per cell).

Regime (Theorem 1’s prediction)	Magnitude	Accuracy [exact 95% CI]	AUC
Confounded (indistinguishable)	0.6	0.529 [0.483, 0.575]	0.519
Separated (distinguishable)	0.6	0.938 [0.912, 0.957]	0.983
Confounded (indistinguishable)	1.2	0.487 [0.442, 0.533]	0.485
Separated (distinguishable)	1.2	1.000 [0.992, 1.000]	1.000

the best risk R_Z^* of a policy of engineered features $Z = \phi(Y)$,

$$R_{\text{oracle}} \leq R_Y^* \leq R_Z^* \leq R_{\text{const}}.$$

The empirical sections estimate R_Z^* , equal to R_Y^* only for sufficient ϕ (not established here). Exact confounding is sufficient, not necessary, for $R_Y^* = R_{\text{const}}$: an informative Y can still leave the optimal action unchanged, for example at a low prior where every posterior stays below the containment threshold. An absence of measured cost benefit is therefore not evidence that two observation laws are equal.

Prior-shift correction. The classifier is trained at a balanced prior, so its score is corrected to the deployment prior by Bayes’ rule before it is compared with $q^* = \text{DISRUPT}/(\text{HARM} - C_{OP} + \text{DISRUPT})$ [36]; without the correction a classifier under confounding can look worse than ignoring the observation (Supplement S4).

Linear pilot. With the illustrative toy triple $(C_{OP}, \text{DISRUPT}, \text{HARM}) = (3, 6, 10)$, which satisfies Proposition 1’s hypotheses, we compare four policies across a grid of priors $P(K)$: always contain, always leave, the prior-corrected classifier, and the oracle.

The constant and oracle policies match their closed forms $pC_{OP} + (1 - p)\text{DISRUPT}$, $p\text{HARM}$ and pC_{OP} , a direct check of the cost computation. Under confounding no policy of Y has expected cost below $R_Y^* = R_{\text{const}}$, so the fitted classifier can at best reach this floor. It did so exactly at every prior below $q^* = 0.462$, taking the best constant action on every test episode in all 20 seeds; at the highest prior, the only one above q^* , its cost scatters around the floor (Supplement S2). Proposition 1 concerns the Bayes-optimal policy; it implies that a fitted classifier cannot beat the floor, not that it attains it. Under separation the classifier beats the best constant policy (Supplement S2, with a sensitivity analysis over cost triples).

Table 2: Linear pilot, confounded regime: expected cost per episode (toy units). Under exact confounding $R_Y^* = R_{\text{const}}$, so the last column is the classifier’s excess over the best achievable cost: mean over seeds, with the range over seeds in brackets. Per-seed values are test-set estimates and can fall on either side of the floor.

$P(K)$	Contain	Leave	Classifier (sd)	Oracle	R_{const}	Classifier $- R_{\text{const}}$ [min, max]
0.05	5.850	0.500	0.500 (0.000)	0.150	0.500	0.000 [0.000, 0.000]
0.1	5.700	1.000	1.000 (0.000)	0.300	1.000	0.000 [0.000, 0.000]
0.2	5.400	2.000	2.000 (0.000)	0.600	2.000	0.000 [0.000, 0.000]
0.3	5.100	3.000	3.000 (0.000)	0.900	3.000	0.000 [0.000, 0.000]
0.5	4.500	5.000	4.520 (0.085)	1.500	4.500	0.020 [-0.125, 0.196]

6.1. Where the dollar cost triple comes from

For the ABIDES study we reconstruct a dollar cost triple $(C_K, C_F, H_K) = (C_{OP}, \text{DISRUPT}, \text{HARM})$ from the Almgren–Chriss market-impact model [37] in its continuous-trading form, the expected shortfall of a uniform-rate liquidation, $L(X, T) = \varepsilon X + (\frac{\gamma}{2} + \frac{\eta}{T})X^2$; the discrete-interval correction it drops, and its size for each window, are in Supplement S4. The triple uses the model’s own illustrative parameters applied to three scenarios: a routine containment trade, a benign order forced into an urgent unwind, and an unmitigated adversarial sequence. This gives $(C_K, C_F, H_K) = (\$490,938, \$473,958, \$1,317,500)$, which satisfies $H_K > C_K$. It is an arithmetic illustration, not a calibration to any venue or incident; the costs are not measured economic losses (worked example in Supplement S4).

6.2. A positive fault loss

The simulated fault mechanisms submit real orders, so leaving a fault alone is not free. We replace the zero entry by H_F , the Almgren–Chriss cost of the fault mechanism’s directly measured submitted volume in each episode, averaged over episodes (so the convexity of L in X is included); it is a modelled cost of submitted, not executed, volume, measured on the same episodes as the classification cells (Section 7.4). Per-design values are in Supplement S4; they range from \$850 to \$104,739 per episode. In D2 the fault’s submitted volume is deterministic and is not changed by the attributes varied, which set the side, price and cancellations of each order but not its size or count, so H_F and the rebaselined C_F are the same in both regimes

(Supplement S4). With $H_F > 0$,

$$R_{\text{oracle}} = p C_K + (1 - p) H_F,$$

$$R(\pi) - R_{\text{oracle}} = p (H_K - C_K) \text{FNR} + (1 - p) (C_F - H_F) \text{FPR},$$

and the cost-optimal threshold on the prior-corrected score is $q^* = (C_F - H_F) / [(H_K - C_K) + (C_F - H_F)]$, which depends on the loss table only (between 0.309 and 0.364 under the current convention and between 0.004 and 0.379 under the rebaselined one; per unit in Supplement S4). In the general table Proposition 1’s floor is $\min(a, b)$, with $a = p(H_K - C_K)$ and $b = (1 - p)(C_F - H_F)$ (Supplement S1, Proposition 2); Proposition 1 is its $H_F = 0$ case. Each cell’s cost is estimated by direct class-conditional weighting, $\hat{R}_p = p\{C_K + (H_K - C_K)\text{FNR}\} + (1 - p)\{H_F + (C_F - H_F)\text{FPR}\}$, from the held-out error rates at q^* , with no resampling step.

Two fault-cost conventions. Both conventions share the compromise column, $C_K = L(X_0, T_u)$ and H_K (Section 6.1), with X_0 the reference order of that section and T_u its urgent horizon, and the same $H_F = \frac{1}{N} \sum_i L(X_i, T_n)$: the cost of each fault episode’s measured volume X_i executed at the pace it actually ran (the post-onset window, T_n). They differ only in C_F .

- *Current convention (primary).* $C_F = \text{DISRUPT} = L(X_0, T_u) - L(X_0, T_p)$, with T_p the patient horizon: the premium of forcing the reference benign order into an urgent rather than a patient unwind, one value for every cell. It models containment that stops the fault’s orders before they execute but disrupts benign business; like H_F , it is an excess cost over the incident-free counterfactual, in which the benign order runs patiently and the fault’s orders do not exist.
- *Rebaselined convention (sensitivity arm).* $C_F = \frac{1}{N} \sum_i L(X_i, T_u)$: containing a fault means unwinding urgently the volume the fault has already put on. Both fault entries are then absolute costs of the same volume, differing only in urgency, so $C_F - H_F$ is an urgency premium, positive because the urgent horizon T_u is shorter than the fixed fault horizon T_n ; C_F varies by design and regime (Supplement S4).

Both conventions satisfy (A7), $H_K > C_K$ and $C_F > H_F$, in every cell (minimum $C_F - H_F$: 369,219 current, 2,914 rebaselined); the formulas above require it. Which convention fits depends on whether containment precedes the fault’s executions: a modelling choice, not a data question. We report

the current convention as primary, as fixed in the analysis plan before the table was computed, and the rebaselined convention as a sensitivity arm.

7. Observation matching in an agent-based market

7.1. Scope and constructions

What this section does and does not test. Theorem 1 concerns equality of the whole observation-trajectory law, $P_K(Y_{0:T}) = P_F(Y_{0:T})$. The constructions below match the two mechanisms on one or a few chosen summary statistics while other observable channels may still differ. Matching one moment is not equality of the joint law, and a classifier exploiting an unmatched channel does not contradict Theorem 1, even in a linear system: $\mathcal{N}((0, 2), I)$ and $\mathcal{N}((0, -2), I)$ share their first-coordinate mean. No linearization bridge (operating point, Jacobian, approximation-error bound) connects the linear model to the simulator. This section is therefore not a test of Theorem 1; it asks which observable channels leak information under which summary-matching choices, and what that does to decision cost.

We inject compromise and fault mechanisms as order attributes (submission rate, order size, price aggressiveness, directional bias, cancel/replace churn) into ABIDES [38], a discrete-event multi-agent market simulator with a background population of noise, value and market-maker agents. Each design has two regimes. In the *summary-matched* regime a calibration search sets the fault’s attributes so that it matches the compromise on a chosen target statistic (the direction of the search is not symmetric; Section 7.2); in the *contrast* regime the two mechanisms are deliberately given a large difference in a chosen attribute. The classifier (logistic regression) uses whole-episode summaries of the observed channels (Section 3.5).

Four designs, five profiles. Design *D1* varies all five attributes and calibrates on mean price drift, with four observed channels (spread, depth, cancellation rate, mid-price). Profile *D1+* reuses *D1*’s episodes with three added channels: order-flow imbalance [39], signed executed volume and queue concentration. *D2* (Study 5 in [40]) varies directional bias, aggressiveness and cancellation rate with a deterministic wake-up clock and calibrates on mean cancellation rate (Supplement S3). *D3* calibrates on total order count and leaves the cancellation rate free. *D4* varies directional bias, size and aggressiveness and calibrates on mean spread. *D1+* is a richer feature set on the same data, not a replication: it shares *D1*’s episodes, fault volume and

H_F , so 50-cell counts include those episodes twice. Text and tables use these codes only.

Era-3 basis. Every headline ABIDES result is generated with the third, hardened version of the simulation pipeline (the *Era-3* pipeline: separate worker processes, one task per worker, and a runtime check that the random-number-generator patches are active; Section 7.4, Supplement S6), on *paired* seeds, each yielding one compromise and one fault episode. Results from the two earlier generation eras are exploratory.

7.2. Classification on the paired basis

Table 3: Classification by design and regime. The ten classification cells are computed on paired seeds: each seed yields one compromise and one fault episode, the train/test split is grouped by seed, and the intervals are cluster-bootstrap intervals over seeds, all under the hardened pipeline (Section 7.4). [†] Zero test errors, so the bootstrap interval is degenerate: accuracy has the exact Clopper–Pearson interval over test seeds (a seed counts as correct when both its episodes are), and AUC the lower bound $\text{TPR}_{\text{lo}} \cdot \text{TNR}_{\text{lo}}$, the product of exact one-sided Clopper–Pearson lower bounds at 98.75% on each rate (joint level 97.5%), valid because $\text{AUC} \geq \text{TPR} \cdot \text{TNR}$ at the fixed threshold (Supplement S6). Other bounds shown as one are rounded.

Design	Contrast		Summary-matched	
	Accuracy [95% CI]	AUC [95% CI]	Accuracy [95% CI]	AUC [95% CI]
D1	0.685 [0.652, 0.718]	0.756 [0.728, 0.786]	0.857 [0.830, 0.882]	0.934 [0.917, 0.950]
D1+	0.998 [0.995, 1.000]	1.0000 [0.9999, 1.0000]	0.898 [0.875, 0.920]	0.966 [0.954, 0.977]
D2	0.663 [0.633, 0.692]	0.731 [0.706, 0.757]	0.553 [0.525, 0.580]	0.577 [0.551, 0.604]
D3	0.998 [0.995, 1.000]	1.0000 [0.9998, 1.0000]	0.875 [0.850, 0.898]	0.953 [0.940, 0.967]
D4	0.727 [0.698, 0.757]	0.807 [0.783, 0.832]	1.000 [0.988, 1.000] [†]	1.000 [0.971, 1] [†]

Every cell has 1000 episodes per class and 600 held-out test episodes from 300 test seeds. A cell is *at chance* when its accuracy interval contains one half, and *above chance* when the lower end of its accuracy interval is above one half; the interval is the cluster bootstrap over test seeds, or the exact Clopper–Pearson interval for a zero-error cell. An interval containing one half would not establish equal observation laws: a wide interval leaves room for undetected signal. Read this way, each cell is a classifier two-sample test [32]: accuracy above one half rejects equality of the two mechanisms’ observation laws. Unlike mimicry in intrusion detection [41], this is no attacker’s search; but the test is symmetric in the labels, so each summary-matched cell above chance shows that its pair is distinguishable, while a compromise tuned to a fixed fault is untested.

Table 3 shows no single ranking. In D1 the summary-matched regime is the more classifiable one, with non-overlapping accuracy intervals; with D1+’s added channels the contrast regime becomes nearly perfectly classifiable, because the contrast mechanisms were always separable on signed executed volume, which the original four channels did not expose (Supplement S3). D2 is weakly classifiable in both regimes; for it and for D1’s summary-matched regime no single leaking channel was identified (the best single-statistic rule of Supplement S4 is descriptive evidence of the dominant channel). In D3 both regimes are strongly classifiable: its mechanisms differ in cancellation rate, one of the observed channels, which matching total order count left free. D4’s summary-matched cell has 0 test errors (exact 95% Clopper–Pearson lower bound 0.988 over test seeds): calibrating on mean spread left a large difference in effective order size, which leaks through depth: a size-based channel, not the cancellation-rate channel of D3. Matching also failed on the target itself: a threshold on mean spread alone beats the best constant policy with a supported interval at every prior but the lowest (Supplement S4), while depth remains the dominant leak. In each of the four injection designs, calibration by search left the summary-matched regime above chance by its interval (Table 3, whose header gives the level): clearly in D1, D3 and D4, and marginally in D2, whose accuracy interval (0.525–0.580) and AUC interval (0.551–0.604) have lower ends close to one half. These are per-cell intervals, not adjusted for the 10 comparisons, and a lower end this close to one half is weak evidence of residual signal: D2’s result is marginal, the other three designs’ are not. With max-T simultaneous intervals over the 10 cells (Supplement S4), D2’s summary-matched accuracy interval is 0.512–0.594 and its AUC interval 0.535–0.619, still above one half, and every summary-matched cell stays above chance. The four designs share one infrastructure (Section 9), so this is a regularity of the constructions tested, not a general law about calibration by search.

Why a matched summary need not reach chance. In the linear model one vector equality in $\mathbb{R}^m$ removes the difference between mechanisms in every observed coordinate at every time step, because C is fixed and known, and even that equality is fragile under state-level injection (Theorem 2). In a multi-agent market there is no fixed linear map to match: a designer matches a finite set of statistics, found by search, and whatever is left unmatched can leak. None of the summary-matched constructions was matched to a pre-specified tolerance on independent data (Supplement S3), so “summary-matched” means a match attempted, not achieved: for D1,

the selected calibration point overshoots its target in every confirmation run (Supplement S3). The cells therefore do not separate leakage through unmatched channels from a residual difference on the target: in D4 a threshold on the target alone beats the best constant policy, and for D2 and D3 the achieved target difference was not estimated.

7.3. Cost of ambiguity on the paired basis

The cost table applies the dollar triple of Section 6.1 and the modelled H_F (from measured submitted volume) of Section 6.2 to five profiles $\times$ two regimes $\times$ five priors (50 cells; full table in Supplement S4). The best constant policy’s excess over the oracle agrees with the identity $B(p) = \min\{p(H_K - C_K), (1 - p)(C_F - H_F)\}$ in every cell. Intervals on differences between policies are cluster-bootstrap intervals over test seeds; the constant policies’ costs are not resampled.

At point estimate (strict inequality), the classifier beats always-contain in 50 of 50 cells and the best constant policy in 44 of 50. With cluster-bootstrap intervals over seeds, its advantage over always-contain is a supported improvement in 49 cells and inconclusive at this power in 1; against the best constant it is a supported improvement in 40 cells and inconclusive in 10. At this sample size no cell shows a supported deterioration against either comparator. No non-inferiority margin was specified, so this is the absence of a detected deterioration, not a non-inferiority result. The 50 cells are not 50 independent comparisons: the five priors of a profile reweight the same test episodes, D1+ re-featurizes D1’s episodes, and the two regimes of a design share seeds, so the cells come from 10 profile–regime units drawn from 8 episode sets. Each verdict uses its own interval, not adjusted for the number of cells; the counts describe the table and carry no family-wise error rate. An adjustment for multiplicity only widens intervals, so it cannot create a supported deterioration; it can only reduce the number of supported improvements. With max-T simultaneous intervals over each family of cells (Supplement S4), the improvement over the best constant is supported in 39 cells under the current convention and 32 under the rebaselined one, no cell shows a supported deterioration, and every unit keeps its class in Table 4. By unit (Table 4), the improvement over the best constant is supported at every prior in 6 units, at some priors in 3, and at none in 1, the D2 summary-matched unit. The cells without a supported improvement (Table 5) lie at the low end of the prior grid or in D2; in all of them the best constant policy

is always-leave, except the D2 summary-matched cell at the highest prior, where it is always-contain.

Table 4: Cost verdicts against the best constant policy by profile–regime unit, under the current and the rebaselined fault-cost conventions: number of priors (of five) with a supported improvement, with a point-estimate exception, and the best constant policy by prior. Per-cell cluster-bootstrap intervals, not adjusted for multiplicity; all cells in Supplement S4.

Design	Regime	Current convention			Rebaselined convention		
		Supported	Exceptions	Best constant	Supported	Exceptions	Best constant
D1	contrast	3	1	LLLLC	3	0	LLCCC
D1	summary-matched	5	0	LLLLC	5	0	LLLLC
D1+	contrast	5	0	LLLLC	5	0	LLCCC
D1+	summary-matched	5	0	LLLLC	5	0	LLLLC
D2	contrast	3	1	LLLLC	0	5	CCCCC
D2	summary-matched	0	4	LLLLC	0	5	CCCCC
D3	contrast	5	0	LLLLC	5	0	CCCCC
D3	summary-matched	5	0	LLLLC	3	1	CCCCC
D4	contrast	4	0	LLLLC	3	0	CCCCC
D4	summary-matched	5	0	LLLLC	5	0	LLCCC

Table 5: Cells whose advantage over the best constant policy is not a supported improvement (paired basis, current fault-cost convention). Costs in dollars (Section 6.1); difference = classifier minus best constant, with its cluster-bootstrap interval over seeds.

Design	Regime	Prior	R_{clf}	Best constant	Difference [95% CI]	vs best constant	vs always-contain
D1	contrast	0.05	99,906	97,510 (leave)	2,395 [−1,791, 7,564]	inconclusive	improvement
D1	contrast	0.10	160,009	161,720 (leave)	−1,711 [−7,827, 5,451]	inconclusive	improvement
D2	contrast	0.05	67,767	66,682 (leave)	1,085 [−827, 4,357]	inconclusive	improvement
D2	contrast	0.10	128,699	132,515 (leave)	−3,816 [−6,894, 401]	inconclusive	improvement
D2	summary-matched	0.05	66,682	66,682 (leave)	0 [0, 0]	inconclusive	improvement
D2	summary-matched	0.10	132,515	132,515 (leave)	0 [0, 0]	inconclusive	improvement
D2	summary-matched	0.20	266,863	264,180 (leave)	2,683 [−551, 7,178]	inconclusive	improvement
D2	summary-matched	0.30	400,582	395,845 (leave)	4,737 [−6,297, 15,234]	inconclusive	improvement
D2	summary-matched	0.50	481,261	482,448 (contain)	−1,187 [−6,308, 4,133]	inconclusive	inconclusive
D4	contrast	0.05	69,353	71,486 (leave)	−2,133 [−5,648, 2,486]	inconclusive	improvement

The exceptions are concentrated in D2 (5 of the 6 point-estimate exceptions); the other profiles have a supported advantage over the best constant in every cell except a few contrast cells at the low end of the prior grid (Table 5). At the lowest priors of D2’s summary-matched regime the classifier’s cost equals the best constant’s exactly: a tie, counted as an exception under strict inequality. Under the rebaselined fault-cost convention (Section 6.2) always-contain is the cheaper constant action in 36 of 50 cells, so the best constant policy changes. The best constant is always-contain exactly at the priors above q^* ; under the rebaselined convention q^* lies below the whole prior grid in D2, D3 and D4’s contrast regime, so there always-contain is the best

constant at every prior. Against it the classifier’s improvement is supported in 34 cells and inconclusive in 16, with no supported deterioration against either comparator. The cells without a supported improvement are no longer confined to the low end of the prior grid and D2 (Table 4), so the location statement above holds under the current convention only (Supplement S4).

Under the current convention, the comparison with always-contain is weak evidence on its own: at every prior of the grid below the highest, always-leave is the cheaper constant action, so any policy that mostly leaves alone beats always-contain. The informative comparison is with the best constant policy.

Practice baseline. Against the best rule thresholding one stored episode statistic, fitted on the training seeds (Supplement S4), the classifier shows no supported deterioration under the current convention; under the rebaselined convention 1 cell is a supported deterioration (D1, contrast regime, prior 0.30; the best rule there thresholds mean spread), inconclusive after a max-T multiplicity adjustment (Supplement S4). There the classifier’s expected cost is 261,875 against the rule’s 250,231 dollars, a difference of 11,644 (per-cell interval from 3,961 to 19,982; max-T interval from $-1,995$ to 25,282). In the summary-matched regime the classifier improves on the rule with a supported interval in D1, D1+ and D3, and is inconclusive in D2; in D4, where it is at ceiling, a depth-variability threshold takes the same actions, so the classifier adds nothing there.

7.4. Checks that bound the claims

Negative control. Two incident variants with identical observable kernels, run through the full pipeline in 20 independent replicates of 30 test episodes each, give mean accuracy 0.505 with 95% interval 0.466–0.544, inside the pre-specified equivalence margin 0.45–0.55; the AUC interval (0.444–0.522) is not, so AUC is inconclusive at this power. The control cannot exclude a pipeline bias up to its interval’s upper end, which lies above the lower end of D2’s summary-matched accuracy interval: a further reason to read D2’s result as marginal (Supplement S5).

Provenance of each result. The ten headline classification cells (Table 3), the cost table built on them and H_F were generated with the hardened pipeline on paired seeds; H_F is computed per fault episode from those same episodes, whose trajectories were recovered and checked bit-identical to the stored features. The multiplicity adjustment, the practice baseline and leave-one-design-out (Supplement S5) reanalyse these episodes; the negative

control is also hardened, with registered seed blocks. The persistence ablation (Supplement S5) and the secondary studies of Supplement S3, except its matching-tolerance estimates and thirty-split check, come from the two earlier generation eras; they are reported as exploratory with their era (Supplement S6), and not all of their seed blocks are in the provenance register.

7.5. Outlook: an action on a second information channel

In the cells of Table 5 (Section 7.3), the fitted classifier’s advantage over the best constant policy is not a supported improvement: passive observation adds no supported value there. Whether an action on a second information channel, such as targeted verification, adds value in those cells is open, and this paper claims no answer; in particular, it does not claim that these cells are where such an action’s effect is largest. The question is addressed, as an exploratory test, in work in preparation [40].

This paper defines no confounded region for a real incident and evaluates no verification channel: confounding (Theorem 1) is a property of the mechanism instances and the observation law, not of the prior p , and only the value of resolving it, Proposition 1’s $\min\{(1-p)\text{DISRUPT}, p(\text{HARM} - C_{OP})\}$, depends on p .

8. Discussion

The linear identifiability boundary is exact within its scope. Its scope is narrower than Theorem 1 alone suggests. Confounding that survives an output-level check need not survive feedback into the dynamics, and the gap is governed by a classical object, the unobservable subspace of (A, C) (Section 4.1). The ABIDES study adds a second, independent caution: the calibration searches of the four designs did not bring any summary-matched regime to chance, and the channels that leaked depended on the design. That calibration needed a search is a fact about the four designs tested, not a dividing line between linear and nonlinear systems. A third caution is methodological: a threshold set without prior-shift correction can manufacture a cost finding (Supplement S4). Supplement S7 turns these observations into a practitioner checklist.

9. Limitations

1. **Retrospective, whole-episode diagnosis.** No result measures the value of acting at a decision deadline (Section 3.5).

2. **One simulator, one shared infrastructure.** The four ABIDES designs are parameterizations of one shared code base (configuration builder, kernel, features, cost pipeline), not independently coded implementations, so a bug common to that infrastructure would not be caught by comparing designs. The five profiles are regularities of the constructions tested; the design space is open-ended. The unit of independent evidence is the design, of which there are four.
3. **Illustrative costs.** The dollar triple is an illustration, not a calibration (Section 6.1), and H_F prices submitted, not executed, volume with a market-impact model; neither is a measured economic loss.
4. **Reference policies.** The constant policies are decision-theoretic reference points, not models of deployed kill switches; the threshold-rule baseline uses stored episode statistics only, so an operational comparison would still need a detector with comparable feature access and timing.
5. **Power and degenerate cells.** Apart from the two lowest-prior D2 summary-matched cells, exact ties in which the fitted policy leaves every test episode alone, the cells of Table 5 are inconclusive at the sample size used, not shown to be ties. The persistence ablation is exploratory (Section 7.4).

10. Conclusion

In a linear-Gaussian model of gateway-observable order flow, the classical criterion makes a compromise and a fault indistinguishable exactly when their projected injections coincide, and the ambiguity then has a strictly positive cost. This confounding is fragile: once the incident propagates through the dynamics, it survives only when the injection difference lies in the Kalman unobservable subspace, which is trivial for almost every dynamics matrix, though structured market dynamics need not be generic (Section 4.1). The criterion extends to random injections (Theorem 4), and for a confounded pair whose state injections differ and whose dynamics are observable, the cost of ambiguity falls below its maximum within at most n post-onset observations under state-level injection, never under output-level injection (Proposition 4).

In the ABIDES simulator, calibration searches that tuned the fault toward a fixed compromise, without reaching a pre-specified match, did not bring the summary-matched regime to chance in any of the four designs, clearly in

three and marginally in D2, also after multiplicity adjustment; which channel leaked depended on the design.

The classifier-based policy shows no supported deterioration against the best constant policy under either fault-cost convention (Section 7.3); Section 7.5 states what the remaining cells leave open.

Data and code availability

The simulation code, the experiment configurations, the provenance register and the result file behind every number in this article are held in the author’s project repository. The hardened-pipeline results (Section 7.4) are listed in the register with their seed blocks; the earlier-era results, reported as exploratory, are deposited as stored outputs, and not all of their seed blocks are registered (Supplement S6). They will be deposited in a public archive on acceptance, with a persistent identifier (DOI to be determined). Until then they are available from the corresponding author on reasonable request. The ABIDES market simulator is open-source software; the patches applied to it are part of the deposit.

CRedit authorship contribution statement

Chiheb Mejrissi: Conceptualization, Methodology, Formal analysis, Software, Investigation, Validation, Data curation, Visualization, Writing – original draft, Writing – review & editing.

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Appendix A. Proofs of the main results

This appendix proves Theorem 2, Corollary 3 and Proposition 4. Supplement S1 proves every numbered statement, one proof per statement, including Theorem 1 and Propositions 1–3, which specialise classical results (Section 2), Theorem 4, and the Remark after Corollary 3. Steps cited below from the proofs of Proposition 3 and Theorem 4 are in Supplement S1.

Appendix A.1. Proof of Theorem 2

Proof. Work in the state-level injection model (Definition D9) with window $W = \{t : t \geq \tau\}$ (Definition D3); by (A4) the onset τ is deterministic and common to both instances (Definition D1). Write $S_k = \sum_{j=0}^k A^j$ for $k \geq 0$.

Step 1 (noise part and deterministic part). Define the uninjected process $z_0 = x_0$, $z_{t+1} = Az_t + w_t$ for all $t \geq 0$, and the deterministic sequence $s_t = 0$ for $t \leq \tau$, $s_{t+1} = As_t + A\theta_M b_M$ for $t \geq \tau$. Then $x_t = z_t + s_t$ for all t : this holds for $t \leq \tau$ because the recursion before the onset is uninjected, and if it holds at some $t \geq \tau$, then by Definition D9

$$x_{t+1} = A(x_t + \theta_M b_M) + w_t = (Az_t + w_t) + (As_t + A\theta_M b_M) = z_{t+1} + s_{t+1}.$$

Unrolling, $s_{\tau+k} = \sum_{j=1}^k A^j \theta_M b_M$, so $\tilde{x}_{\tau+k} = z_{\tau+k} + S_k \theta_M b_M$ and, for every $k \geq 0$,

$$y_{\tau+k} = G_{\tau+k} + m_M(\tau+k), \quad G_t = Cz_t + \varepsilon_t, \quad m_M(\tau+k) = CS_k \theta_M b_M.$$

Step 2 (why the noise does not matter). $G = (G_t)_{t \geq 0}$ is a fixed measurable function of $(x_0, \{w_t\}, \{\varepsilon_t\})$ through A and C and does not involve τ or the instance. By (A1), (A2) and (A3) its law, in particular its covariance, is the same under both instances. The instance enters only through the deterministic mean shift m_M .

Step 3 (laws equal iff shifts equal). If $m_K(t) = m_F(t)$ for all $t \in W$, then $Y_W = G_W + m$ under both instances, so every finite-dimensional marginal of Y_W is the same translate of the common marginal of G , and $P_K^W = P_F^W$ (Definition D3). Conversely, if $P_K^W = P_F^W$, then since each G_t is Gaussian by (A1) and (A2), and hence integrable, $\mathbb{E}_K[y_t] = \mathbb{E}_F[y_t]$ and so $m_K(t) = m_F(t)$

for every $t \in W$. With $\Delta_0 = \theta_K b_K - \theta_F b_F$, $m_K(\tau + k) - m_F(\tau + k) = CS_k \Delta_0$. Hence

$$P_K^W = P_F^W \iff CS_k \Delta_0 = 0 \text{ for all } k \geq 0.$$

Step 4 (telescoping). For every $N \geq 0$: $CS_k \Delta_0 = 0$ for $k = 0, \dots, N \iff CA^k \Delta_0 = 0$ for $k = 0, \dots, N$. Indeed $CS_0 = C$ and $CS_k - CS_{k-1} = CA^k$ for $k \geq 1$, so each family is obtained from the other by taking successive differences or partial sums. Hence $P_K^W = P_F^W \iff CA^k \Delta_0 = 0$ for all $k \geq 0$.

Step 5 (Cayley–Hamilton). Let $\chi(\lambda) = \lambda^n + c_{n-1}\lambda^{n-1} + \dots + c_0$ be the characteristic polynomial of A . By the Cayley–Hamilton theorem $A^n = -\sum_{i=0}^{n-1} c_i A^i$; multiplying by A^{k-n} and inducting on k , every A^k with $k \geq n$ is a linear combination $\sum_{i=0}^{n-1} \alpha_{k,i} A^i$. Therefore, if $CA^i \Delta_0 = 0$ for $i = 0, \dots, n-1$, then $CA^k \Delta_0 = \sum_i \alpha_{k,i} CA^i \Delta_0 = 0$ for every $k \geq n$; the converse is immediate. So

$$P_K^W = P_F^W \iff \Delta_0 \in \bigcap_{k=0}^{n-1} \ker(CA^k) = \mathcal{N}(A, C)$$

(Definition D10), which is the claim.

Remark. Adding pre-onset times to W changes nothing, since $m_K(t) = m_F(t) = 0$ for $t < \tau$. $\square$

Appendix A.2. Proof of Corollary 3

Proof. By Theorem 2 (Definition D9, Definition D1, Definition D3, (A1)–(A4), (A5) in its all-time form), the instances are indistinguishable at every $t \geq \tau$ iff $\Delta_0 = \theta_K b_K - \theta_F b_F \in \mathcal{N}(A, C)$. If (A, C) is observable, $\mathcal{N}(A, C) = \{0\}$ (Definition D10), so the condition is $\Delta_0 = 0$, i.e. $\theta_K b_K = \theta_F b_F$. $\square$

Appendix A.3. Proof of Proposition 4

Proof. Step 1 (the two laws on W_H). With $Z = Y_{W_H}^0$ the injection-free observation on W_H (Step 0 of the proof of Theorem 4, deterministic case $s_M = \theta_M b_M$), $Y_{W_H} = Z + G_H \theta_M b_M$ under state-level injection and $Y_{W_H} = Z + \mathbf{1}_H \otimes C \theta_M b_M$ under output-level injection (every $t \in W_H$ is post-onset). $Z \sim \mathcal{N}(\mu^0, V_H)$ does not depend on the cause or on the model, so V_H is common. By (A1) and (A2), ε is independent of x^0 , so $V_H = \text{Cov}((Cx_t^0)_{t \in W_H}) + I_H \otimes \Sigma \succ 0$ by (A6). Hence $Y_{W_H} \mid M \sim \mathcal{N}(\mu_M, V_H)$ with $\mu_K - \mu_F = \delta_H$, where $\delta_H = G_H \Delta_0$ (state level) or $\delta_H = \mathbf{1}_H \otimes C \Delta_0$ (output level), and $d_H = \|V_H^{-1/2} \delta_H\|$ is the discriminability of Definition D5.

Step 2 (closed form). If $d_H > 0$, Proposition 3 applies to $Y = Y_{W_H}$ and gives $\text{CoA}_H^* = \text{CoA}_Y^*(d_H)$. If $d_H = 0$, then $\delta_H = 0$ ($V_H \succ 0$), the two laws coincide, and Step 1 of the proof of Proposition 3 gives $\text{CoA}_H^* = \int \min\{af, bf\} dy = \min(a, b)$.

Step 3 (a strict inequality). By Proposition 3, $g(d) = \text{CoA}_Y^*(d)$ is strictly decreasing on $(0, \infty)$ with $g(d) \rightarrow \min(a, b)$ as $d \rightarrow 0^+$ (its proof, Steps 4–5). Hence $g(d) < \min(a, b)$ for every $d > 0$. With Step 2: $\text{CoA}_H^* < \min(a, b)$ iff $d_H > 0$ iff $\delta_H \neq 0$.

(i). $W_H \subset W_{H+1}$, so every policy measurable in Y_{W_H} is a policy on W_{H+1} with the same expected cost (Definition D7); the infimum over the larger class is no larger, $R_{H+1}^* \leq R_H^*$, while R_{oracle} does not depend on H . Hence $\text{CoA}_{H+1}^* \leq \text{CoA}_H^*$. (Equivalently, d_H is non-decreasing: δ_H and V_H are the leading sub-vector and block of δ_{H+1} and V_{H+1} , and with S the Schur complement of V_H in V_{H+1} , $S \succ 0$, $\delta_{H+1}^\top V_{H+1}^{-1} \delta_{H+1} = \delta_H^\top V_H^{-1} \delta_H + r^\top S^{-1} r \geq \delta_H^\top V_H^{-1} \delta_H$ for the appropriate residual r .)

(ii). Under output-level injection $\delta_H = \mathbf{1}_H \otimes C\Delta_0$ is zero iff $C\Delta_0 = 0$, for every H . Step 2 and Step 3 conclude.

(iii). Under state-level injection $\delta_H = G_H\Delta_0$, which is zero iff $CA^j\Delta_0 = 0$ for $j = 0, \dots, H-1$ (Theorem 3). Step 3 gives the equivalence. If (A, C) is observable and $\Delta_0 \neq 0$, then $\Delta_0 \notin \bigcap_{k=0}^{n-1} \ker(CA^k) = \{0\}$, so $j^* = \min\{j : CA^j\Delta_0 \neq 0\}$ exists and $j^* \leq n-1$; for every $H \geq h^* = j^* + 1 \leq n$, $j^* < H$ and the condition holds.

Final sentence. If $C\Delta_0 = 0$, (ii) gives $\min(a, b)$ at every horizon under output-level injection; under state-level injection $j^* \geq 1$ because $CA^0\Delta_0 = C\Delta_0 = 0$, so $h^* \in \{2, \dots, n\}$, and (iii) gives $\text{CoA}_H^* < \min(a, b)$ for every $H \geq h^*$. $\square$

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